

Two-way fixed effects estimators with heterogeneous treatment effects*

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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This result implies that in general, $\beta_{fe} \neq \delta^{TR}$, so $\hat{\beta}_{fe}$ is a biased estimator of the ATT. To illustrate this, we consider a simple example of a staggered adoption design with two groups and three periods, and where the treatments are non-stochastic: group 1 is untreated at periods 1 and 2 and treated at period 3, while group 2 is untreated at period 1 and treated both at periods 2 and 3.⁸ We also assume that $N_{g,t}/N_{g,t-1}$ does not vary across g : all groups experience the same growth of their number of observations from $t - 1$ to t , a requirement that is for instance satisfied when the data is a balanced panel. Then, one can show that

$$\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot},$$

thus implying that

$$\begin{aligned}\varepsilon_{1,3} &= 1 - 1/3 - 1 + 1/2 = 1/6, \\ \varepsilon_{2,2} &= 1 - 2/3 - 1/2 + 1/2 = 1/3, \\ \varepsilon_{2,3} &= 1 - 2/3 - 1 + 1/2 = -1/6.\end{aligned}$$

The residual is negative in group 2 and period 3, because the regression predicts a treatment probability larger than one in that cell, a classic extrapolation problem with linear regressions. Then, under the common trends assumption, it follows from Theorem 1 and the fact that the treatments are non-stochastic that

$$\beta_{fe} = 1/2E[\Delta_{1,3}] + E[\Delta_{2,2}] - 1/2E[\Delta_{2,3}].$$

β_{fe} is equal to a weighted sum of the ATEs in group 1 at period 3, group 2 at period 2, and group 2 at period 3, the three treated (g, t) cells. However, the weight assigned to each ATE differs from $1/3$, the proportion that each cell accounts for in the population of treated observations. Therefore, β_{fe} is not equal to δ^{TR} . Perhaps more worryingly, not all the weights are positive: the weight assigned to the ATE in group 2 period 3 is strictly negative. Consequently, β_{fe} may be a very misleading measure of the treatment effect. Assume for instance that $E[\Delta_{1,3}] = E[\Delta_{2,2}] = 1$ and $E[\Delta_{2,3}] = 4$. At the period when they start receiving the treatment, both groups experience a modest positive ATE. But this effect builds over time and in period 3, one period after it has started receiving the treatment, group 2 now experiences a large ATE. Then,

$$\beta_{fe} = 1/2 \times 1 + 1 - 1/2 \times 4 = -1/2.$$

β_{fe} is strictly negative, while $E[\Delta_{1,3}]$, $E[\Delta_{2,2}]$, and $E[\Delta_{2,3}]$ are all positive. More generally, the negative weights are an issue if the $E[\Delta_{g,t}]$ s are heterogeneous, across groups or over time.⁹ If $E[\Delta_{1,3}] = E[\Delta_{2,2}] = E[\Delta_{2,3}] = 1$, then $\beta_{fe} = 1 = \delta^{TR}$.

⁸ A similar example appears in Borusyak and Jaravel (2017).

⁹ On the other hand, β_{fe} does not rule out heterogeneous treatment effects within (g, t) cells, as it is identified by variations across (g, t) cells, and does not leverage any within-cell variation.